

The Exact Zero-Sum Ramsey Numbers of Disjoint Triangles Modulo 3

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Abstract

For a simple graph G with $k \mid e(G)$, the zero-sum Ramsey number $R(G, \mathbb{Z}_k)$ is the smallest positive integer N such that every edge-weighting $c : E(K_N) \rightarrow \mathbb{Z}_k$ contains a zero-sum copy of G . Addressing Problem 4 posed by Caro and Mifsud (*arXiv:2502.03864*), which asks whether $R(G, \mathbb{Z}_3) \leq |V(G)| + 8$ with equality if and only if $G = K_3$, we prove the exact value for any collection of $s \geq 2$ vertex-disjoint triangles:

$$R(sK_3, \mathbb{Z}_3) = 3s + 2 \quad \text{for all } s \geq 2.$$

The lower bound $R(sK_3, \mathbb{Z}_3) \geq 3s + 2$ is derived analytically for all $s \geq 1$ via a vertex-potential construction on K_{3s+1} . The upper bound is established by coupling a propositional refutation of K_8 (252 variables, 2,689 clauses) certified via CDCL SAT solvers with an inductive reduction that peels off zero-sum triangles using $R(K_3, \mathbb{Z}_3) = 11$. Consequently, the additive surplus parameter $\Delta(G) := R(G, \mathbb{Z}_3) - |V(G)|$ collapses from $\Delta(K_3) = +8$ to $\Delta(sK_3) = +2$ for all $s \geq 2$, matching the known universal upper bound for acyclic forests.

1 Introduction

Let $G = (V, E)$ be a finite simple graph, and let $\mathbb{Z}_k$ denote the cyclic group of order k . An edge-weighting of the complete graph K_N is a map $c : E(K_N) \rightarrow \mathbb{Z}_k$. For any subgraph $H \subseteq K_N$, its total edge weight is defined by

$$w(H) := \sum_{e \in E(H)} c(e) \pmod{k}.$$

The subgraph H is called *zero-sum* if $w(H) \equiv 0 \pmod{k}$. The *zero-sum Ramsey number* $R(G, \mathbb{Z}_k)$ is the smallest positive integer N such that every edge-weighting $c : E(K_N) \rightarrow \mathbb{Z}_k$ forces a zero-sum subgraph isomorphic to G . The parameter $R(G, \mathbb{Z}_k)$ exists if and only if $k \mid e(G)$; otherwise, assigning the constant weight $c(e) = 1$ leaves no zero-sum subgraph of any order.

Zero-sum Ramsey theory was introduced by Bialostocki and Dierker [1] as an extension of classical Ramsey theory and the Erdős-Ginzburg-Ziv theorem [5]. In 1996, Caro [2] completely resolved $R(G, \mathbb{Z}_2)$ for all graphs with even edge counts.

Over $\mathbb{Z}_3$, zero-sum Ramsey numbers display a sharp threshold difference between acyclic and dense cyclic graphs. For every forest F on n vertices with $3 \mid e(F)$, Caro and Mifsud [3] proved:

$$R(F, \mathbb{Z}_3) \leq n + 2. \tag{1}$$

In stark contrast, for the single triangle K_3 ($n = 3, e = 3$), classical evaluations establish:

$$R(K_3, \mathbb{Z}_3) = 11 = |V(K_3)| + 8 \tag{2}$$

(see Chung and Graham [4], and Harborth and Piepmeyer [6]). Defining the *additive surplus parameter* by

$$\Delta(G) := R(G, \mathbb{Z}_3) - |V(G)|,$$

forests satisfy $\Delta(F) \leq 2$, whereas the single triangle exhibits the anomalous surplus $\Delta(K_3) = 8$. This motivated the following question:

Problem 1.1 (Caro & Mifsud [3], Problem 4). Let G be a graph on n vertices such that $3 \mid e(G)$. Is it true that

$$R(G, \mathbb{Z}_3) \leq n + 8,$$

with equality if and only if $G = K_3$?

In this paper, we resolve the exact value for the disjoint union of $s \geq 2$ triangles sK_3 ($|V(sK_3)| = 3s$, $e(sK_3) = 3s$).

Theorem 1.2. *For every integer $s \geq 2$, the zero-sum Ramsey number of s vertex-disjoint triangles modulo 3 is*

$$R(sK_3, \mathbb{Z}_3) = 3s + 2.$$

In particular, $\Delta(sK_3) = 2$ for all $s \geq 2$.

Theorem 1.2 confirms that the extreme surplus $\Delta(K_3) = 8$ is strictly confined to the isolated triangle: the moment a second disjoint component is introduced ($s \geq 2$), the surplus collapses to +2, matching the universal forest boundary.

2 The Analytical Lower Bound: $R(sK_3, \mathbb{Z}_3) \geq 3s + 2$

Proposition 2.1. *For every integer $s \geq 1$:*

$$R(sK_3, \mathbb{Z}_3) \geq 3s + 2.$$

Proof. Let $N = 3s + 1$. We construct an edge-weighting $c : E(K_N) \rightarrow \mathbb{Z}_3$ containing no zero-sum copy of sK_3 via the vertex-potential method.

Assign a potential value $a_v \in \mathbb{Z}_3$ to each vertex $v \in V(K_N)$, and define the edge weights by

$$c(uv) := a_u + a_v \pmod{3}.$$

For any subgraph $H \subseteq K_N$, its total weight satisfies:

$$w(H) = \sum_{uv \in E(H)} (a_u + a_v) = \sum_{v \in V(H)} \deg_H(v) a_v \pmod{3}.$$

In any copy of sK_3 , every vertex $v \in V(sK_3)$ belongs to a triangle component, and thus has degree $\deg(v) = 2 \equiv -1 \pmod{3}$. Therefore:

$$w(sK_3) \equiv \sum_{v \in V(sK_3)} (-1) a_v = - \sum_{v \in V(sK_3)} a_v \pmod{3}.$$

Because sK_3 spans exactly $3s$ vertices and $|V(K_N)| = N = 3s + 1$, every copy of sK_3 in K_N omits exactly one vertex $w \in V(K_N)$. Let

$$S := \sum_{v \in V(K_N)} a_v \pmod{3}$$

denote the total potential sum of the host graph. The weight of any copy of sK_3 in K_N then satisfies:

$$w(sK_3) \equiv -(S - a_w) \equiv a_w - S \pmod{3}. \quad (3)$$

We define the vertex potentials $a : V(K_N) \rightarrow \{1, 2\}$ explicitly:

- Exactly 2 vertices have potential 2;
- The remaining $N - 2 = (3s + 1) - 2 = 3s - 1$ vertices have potential 1.

The total potential sum is:

$$S = 2(2) + (3s - 1)(1) = 3s + 3 \equiv 0 \pmod{3}.$$

Substituting $S \equiv 0 \pmod{3}$ into equation (3), we obtain:

$$w(sK_3) \equiv a_w \pmod{3}.$$

Because every vertex in K_N has potential $a_v \in \{1, 2\}$, the omitted vertex w necessarily satisfies $a_w \not\equiv 0 \pmod{3}$. Consequently:

$$w(sK_3) \not\equiv 0 \pmod{3}$$

for every copy of sK_3 in K_N . Thus, K_{3s+1} admits an edge-weighting with no zero-sum sK_3 , establishing $R(sK_3, \mathbb{Z}_3) \geq 3s + 2$. $\square$

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3 The Base Case: $R(2K_3, \mathbb{Z}_3) \leq 8$

To initialize induction, we establish the base case $s = 2$. Because the single-triangle threshold is $R(K_3, \mathbb{Z}_3) = 11 > 8$, one cannot deduce the existence of a zero-sum triangle inside K_8 directly. We resolve this step via propositional satisfiability.

Proposition 3.1. *Every edge-weighting $c : E(K_8) \rightarrow \mathbb{Z}_3$ contains a zero-sum copy of $2K_3$. Consequently, $R(2K_3, \mathbb{Z}_3) \leq 8$.*

Proof. We encode the existence of a zero-sum-free edge-weighting of K_8 as an exact Conjunctive Normal Form (CNF) formula:

1. Variables:

- For each of the $\binom{8}{2} = 28$ edges $e \in E(K_8)$, three boolean variables $x_{e,0}, x_{e,1}, x_{e,2}$ encode its weight in $\mathbb{Z}_3$ (84 variables).
- For each of the $\binom{8}{3} = 56$ triangles $T \in \binom{V(K_8)}{3}$, three boolean variables $r_{T,0}, r_{T,1}, r_{T,2}$ encode its residue modulo 3 (168 variables).

Total variables: $84 + 168 = 252$.

2. Clauses:

- *Edge exact-one constraints:* Each edge receives exactly one weight: $28 \times 4 = 112$ clauses.
- *Triangle exact-one constraints:* Each triangle receives exactly one residue: $56 \times 4 = 224$ clauses.

- *Triangle definitions:* For each triangle T with edges e_1, e_2, e_3 and each triple $(c_1, c_2, c_3) \in \{0, 1, 2\}^3$, we include the implication clause $\neg x_{e_1, c_1} \vee \neg x_{e_2, c_2} \vee \neg x_{e_3, c_3} \vee r_{T, (c_1+c_2+c_3 \bmod 3)}$: $56 \times 27 = 1,512$ clauses.
- *Zero-sum avoidance:* There are $\frac{1}{2} \binom{8}{3} \binom{5}{3} = 280$ unordered pairs of vertex-disjoint triangles $\{T_1, T_2\}$. A disjoint pair forms a zero-sum $2K_3$ if and only if $(w(T_1), w(T_2)) \in \{(0, 0), (1, 2), (2, 1)\}$. Forbidding these combinations across all 280 pairs contributes $280 \times 3 = 840$ clauses.
- *Symmetry breaking:* Fixing $x_{(0,1),0} = \text{True}$ preserves satisfiability without loss of generality (1 clause).

Total clause count:

$$112 + 224 + 1,512 + 840 + 1 = 2,689.$$

This exact 2,689-clause instance was evaluated using CDCL SAT solvers (including Z3 and CaDiCaL). The formula is determined to be **UNSATISFIABLE** in under 0.2 seconds. Hence, no zero-sum-free edge-weighting exists on K_8 , establishing $R(2K_3, \mathbb{Z}_3) \leq 8$. $\square$

Combining Proposition 2.1 (for $s = 2$) and Proposition 3.1 yields the exact base value:

Corollary 3.2. $R(2K_3, \mathbb{Z}_3) = 8$.

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4 The Inductive Step and Proof of Theorem 1.2

With the base case certified, we establish the matching upper bound for all $s \geq 2$.

Proposition 4.1. *For every integer $s \geq 2$:*

$$R(sK_3, \mathbb{Z}_3) \leq 3s + 2.$$

Proof. We proceed by induction on $s \geq 2$.

Base step ($s = 2$): By Proposition 3.1, $R(2K_3, \mathbb{Z}_3) \leq 8 = 3(2) + 2$.

Inductive step: Assume that $R(sK_3, \mathbb{Z}_3) \leq 3s + 2$ holds for some integer $s \geq 2$. We consider $(s + 1)$ disjoint triangles and set

$$N := 3(s + 1) + 2 = 3s + 5.$$

Let $c : E(K_N) \rightarrow \mathbb{Z}_3$ be an arbitrary edge-weighting of K_N . Because $s \geq 2$, the order of K_N satisfies:

$$N = 3s + 5 \geq 3(2) + 5 = 11.$$

By the classical evaluation $R(K_3, \mathbb{Z}_3) = 11$ [4, 6], every edge-weighting of K_{11} contains at least one zero-sum triangle. Since $N \geq 11$, the host graph K_N contains a zero-sum triangle T_0 :

$$w(T_0) \equiv 0 \pmod{3}.$$

Remove the 3 vertices of T_0 from K_N and define the induced complete subgraph

$$K' := K_N \setminus V(T_0).$$

The number of vertices in K' is:

$$|V(K')| = N - 3 = (3s + 5) - 3 = 3s + 2.$$

By our induction hypothesis, $R(sK_3, \mathbb{Z}_3) \leq 3s + 2$. Therefore, K' contains a zero-sum copy of sK_3 , denoted H :

$$w(H) \equiv 0 \pmod{3}.$$

Because $V(H) \subseteq V(K') = V(K_N) \setminus V(T_0)$, the triangle T_0 is vertex-disjoint from all s triangle components of H . Consequently, the union

$$H^* := T_0 \cup H$$

is a subgraph of K_N isomorphic to $(s + 1)K_3$. Its total weight satisfies:

$$w(H^*) = w(T_0) + w(H) \equiv 0 + 0 \equiv 0 \pmod{3}.$$

Thus, H^* is a zero-sum copy of $(s + 1)K_3$ in K_N . This proves $R((s + 1)K_3, \mathbb{Z}_3) \leq 3(s + 1) + 2$, completing the induction. $\square$

Proof of Theorem 1.2. Combining Proposition 2.1 and Proposition 4.1 yields

$$3s + 2 \leq R(sK_3, \mathbb{Z}_3) \leq 3s + 2 \implies R(sK_3, \mathbb{Z}_3) = 3s + 2$$

for all $s \geq 2$. The additive surplus is $\Delta(sK_3) = (3s + 2) - 3s = 2$. $\square$

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Reproducibility and Code Availability

The propositional formula generator, DIMACS CNF file, and automated CDCL verification script are publicly available at <https://github.com/vegaorion/zero-sum-ramsey-2k3>.

References

- [1] A. Bialostocki and P. Dierker, *Zero-sum Ramsey theorems*, Congr. Numer. **89** (1992), 193–198.
- [2] Y. Caro, *Zero-sum problems—a survey*, Discrete Math. **152** (1996), no. 1-3, 93–113.
- [3] Y. Caro and X. Mifsud, *On zero-sum Ramsey numbers modulo 3*, arXiv preprint arXiv:2502.03864 (2025).
- [4] F. R. K. Chung and R. L. Graham, *Zero-sum problems in graph theory*, Sympos. Gaussian, Conf. A, Math. Hist. (1992), 133–150.
- [5] P. Erdős, A. Ginzburg, and A. Ziv, *Theorem in the additive number theory*, Bull. Res. Council Israel Sect. F **10** (1961), 41–43.
- [6] H. Harborth and L. Piepmeyer, *R-numbers for complete graphs with modulo conditions*, Discrete Math. **113** (1993), 263–268.